Lab Title:- ILLEGAL FILE TRANSFER(MEMORY FORENSICS)

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Course Name:- COMPUTER FORENSICS CASE STUDIES

Instructor Name:- COMMANDANT AMINU IDRIS

Date Of Submission:- 19/10/2026

Case scenario

Investigators obtained a Windows 10 memory image from a workstation after a confidential document may have been transferred without authorization. The inquiry is whether browser activity, a command prompt, a removable device, and a user profile form a defensible sequence. This is an offline examination of the supplied material. It does not authorize contact with any address, host, account, file-sharing service, or person that may appear in the image.

Lab Objectives

The objectives of the lab is to understand how to use volatility to analyse a memory image.

Tools and Resources Used

- Kali linux:- A penetration testing platform used for security assessments.

Kali linux tools

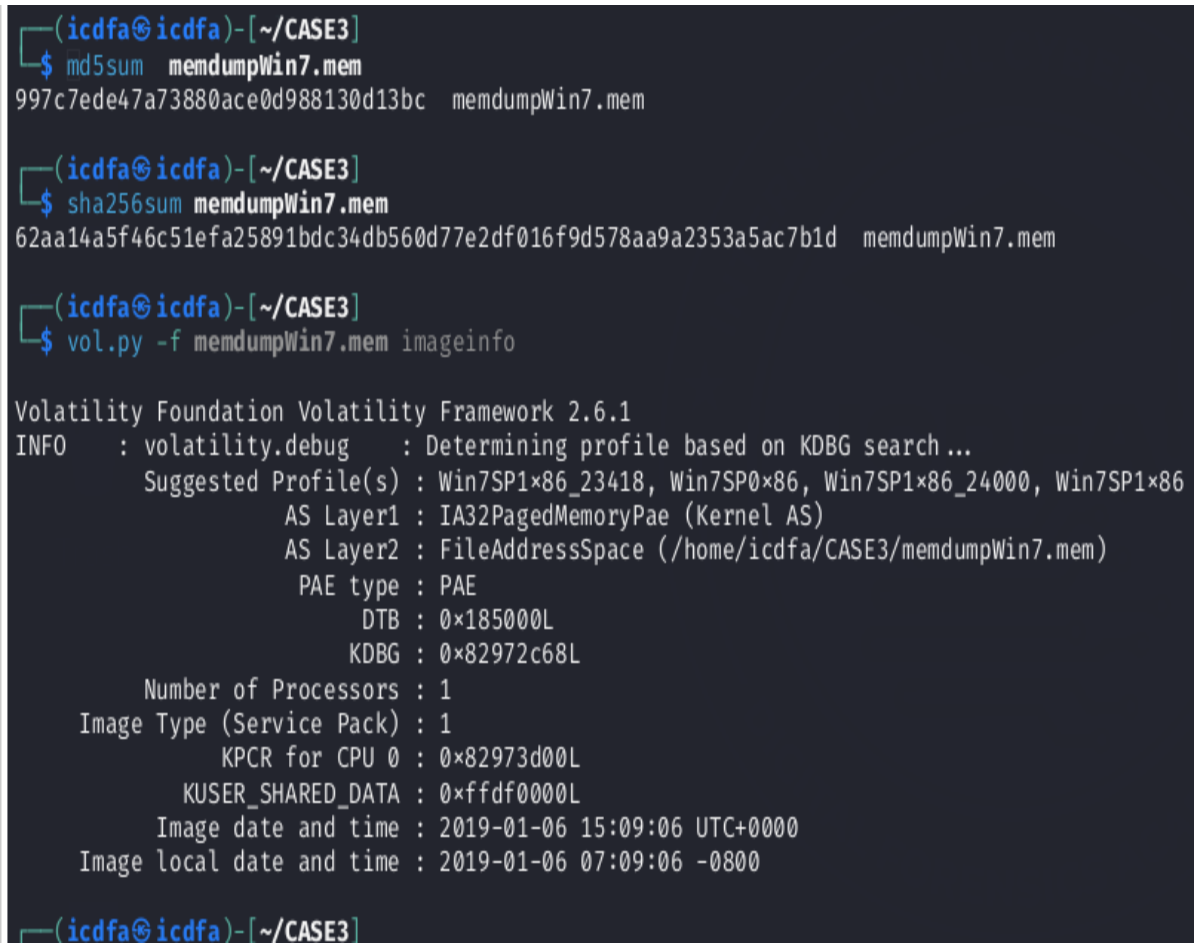
- Volatility(vol.py):-is a platform for memory forensic analysis.

Methology

- I power on the kali linux machine
- I used the wget command to download the memory forensic file(memdumpWin7.mem)
- I obtain some basic information about the memory forensic file(memdumpWin7.mem) such as the MD5 & SHA256 hash values
- I start analyzing the memory forensics file(memdumpWin7.mem) using the volatility tool with different options.

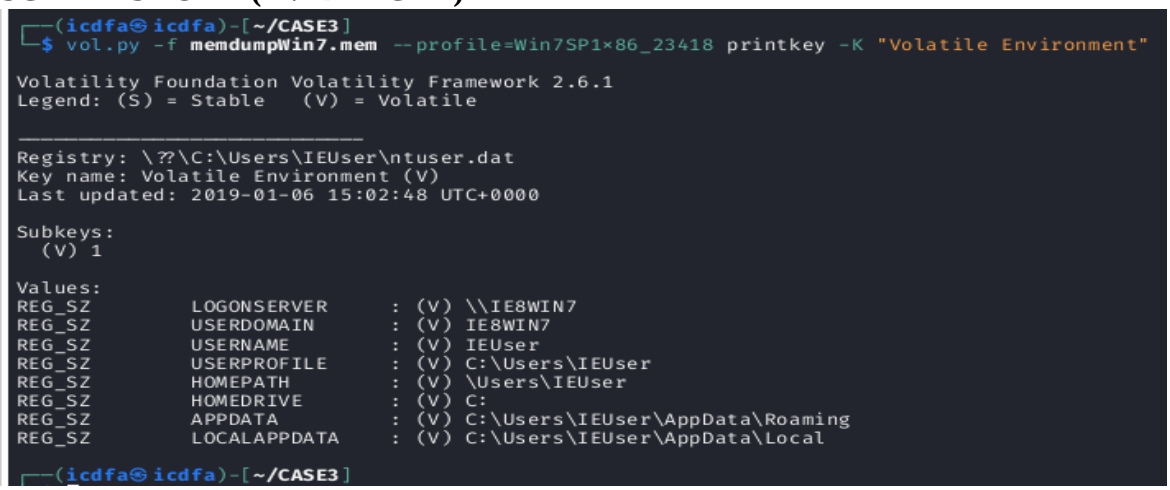
Screenshot and Evidence

SCREENSHOT SHOWING SOME BASIC INFORMATION ABOUT THE HISTORY FILE

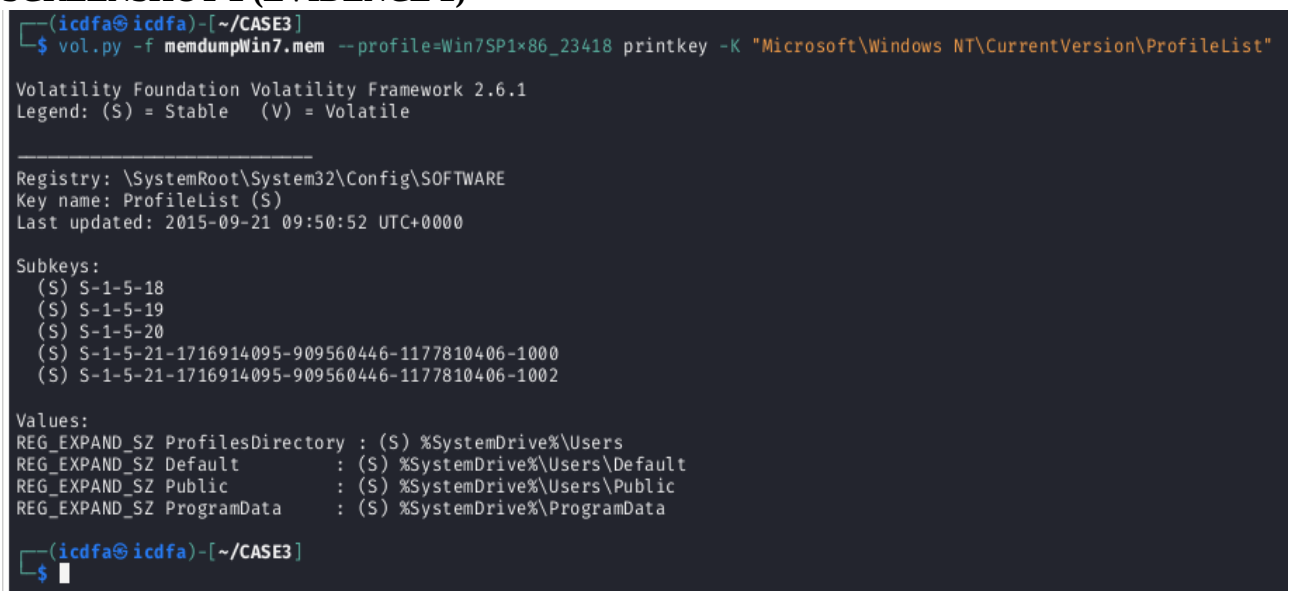


UNDERSTANDING THE SUSPECT AND ACCOUNTS

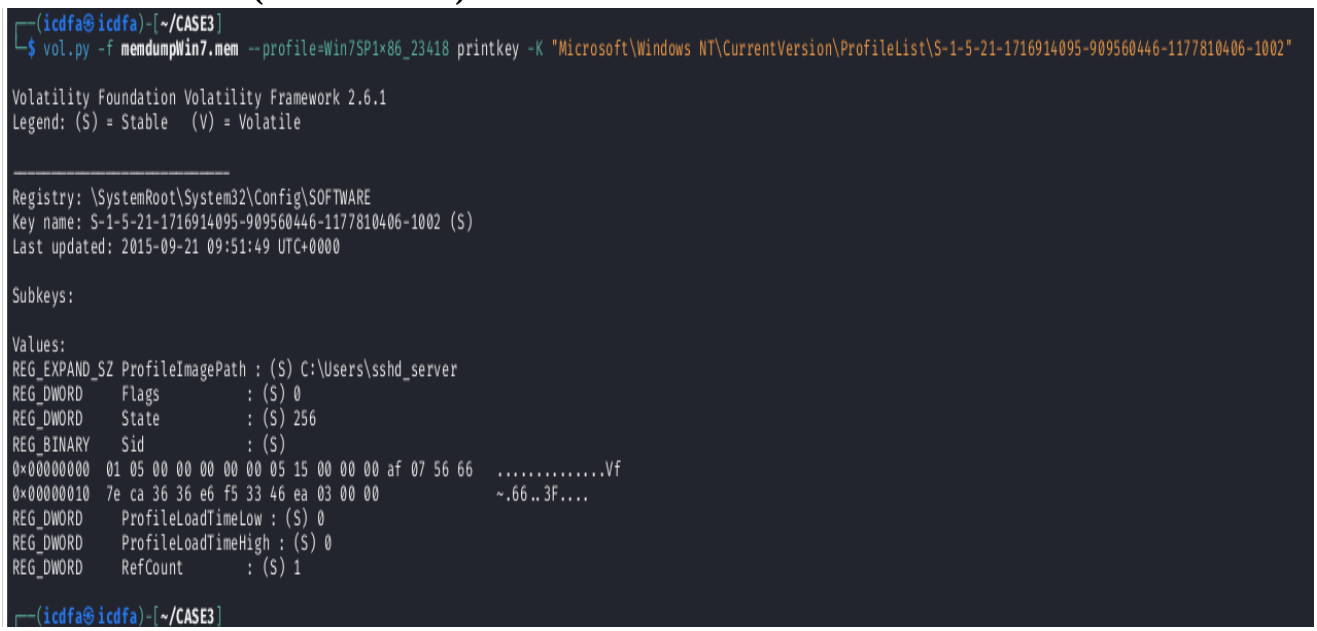
SCREENSHOT 1 (EVIDENCE 1)



SCREENSHOT 2 (EVIDENCE 2)



SCREENSHOT 3 (EVIDENCE 3)



SCREENSHOT 4 (EVIDENCE 4)



SCREENSHOT 5 (EVIDENCE 5)

```
┌──(icdfa㉿icdfa)-[~/CASE3]
└─$ vol.py -f memdumpWin7.mem --profile=Win7SP1x86_23418 printkey -K "Microsoft\Windows NT\CurrentVersion\Winlogon"
Volatility Foundation Volatility Framework 2.6.1
Legend: (S) = Stable   (V) = Volatile

----------------------------
Registry: \SystemRoot\System32\Config\SOFTWARE
Key name: Winlogon (S)
Last updated: 2019-01-06 15:02:47 UTC+0000

Subkeys:
  (S) GPExtensions
  (V) AutoLogonChecked

Values:
REG_SZ        ReportBootOk    : (S) 1
REG_SZ        Shell           : (S) explorer.exe
REG_SZ        PreCreateKnownFolders : (S) {A520A1A4-1780-4FF6-BD18-167343C5AF16}
REG_SZ        Userinit        : (S) C:\Windows\system32\userinit.exe,
REG_SZ        VMApplet        : (S) SystemPropertiesPerformance.exe /pagefile
REG_SZ        WinStationsDisabled : (S) 0
REG_DWORD     DisableCAD      : (S) 1
REG_SZ        scremoveoption  : (S) 0
REG_DWORD     ShutdownFlags   : (S) 2147483687
REG_SZ        DefaultDomainName : (S)
REG_SZ        DefaultUserName : (S) IEUser
REG_SZ        AutoAdminLogon  : (S) 1

┌──(icdfa㉿icdfa)-[~/CASE3]
└─$
```

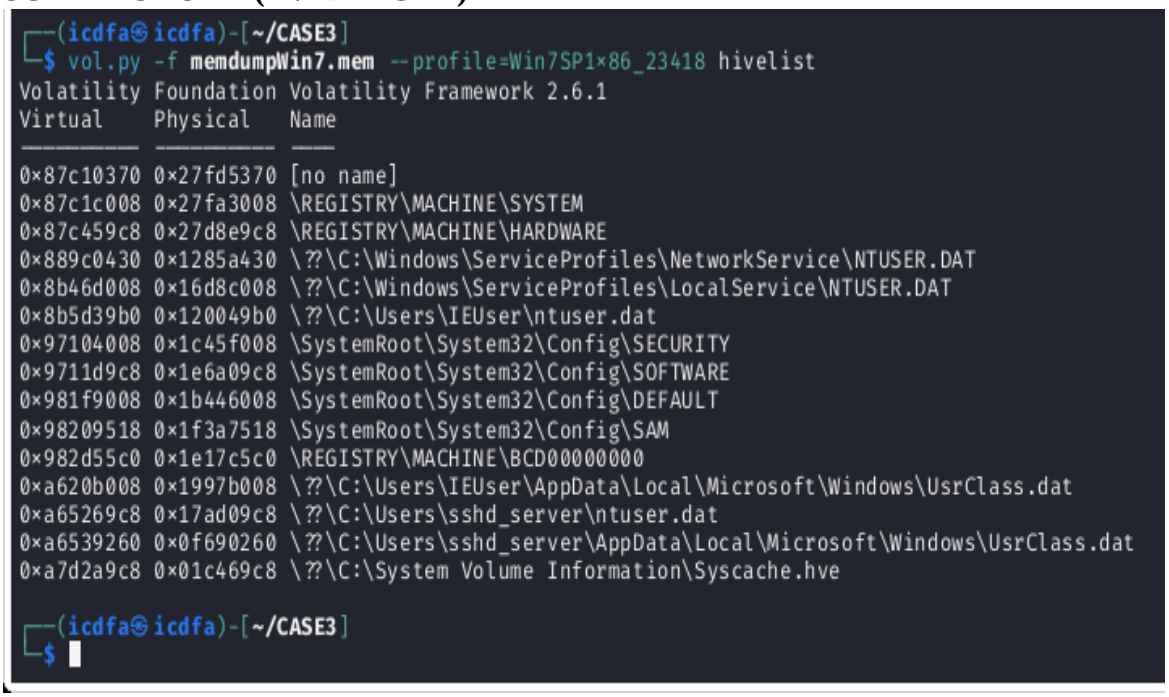
SCREENSHOT 6 (EVIDENCE 6

```
┌──(icdfa㉿icdfa)-[~/CASE3]
└─$ vol.py -f memdumpWin7.mem --profile=Win7SP1x86_23418 hivelist
Volatility Foundation Volatility Framework 2.6.1
Virtual    Physical   Name
---------- ---------- ----
0x87c10370 0x27fd5370 [no name]
0x87c1c008 0x27fa3008 \REGISTRY\MACHINE\SYSTEM
0x87c459c8 0x27d8e9c8 \REGISTRY\MACHINE\HARDWARE
0x889c0430 0x1285a430 \??\C:\Windows\ServiceProfiles\NetworkService\NTUSER.DAT
0x8b46d008 0x16d8c008 \??\C:\Windows\ServiceProfiles\LocalService\NTUSER.DAT
0x8b5d39b0 0x120049b0 \??\C:\Users\IEUser\ntuser.dat
0x97104008 0x1c45f008 \SystemRoot\System32\Config\SECURITY
0x9711d9c8 0x1e6a09c8 \SystemRoot\System32\Config\SOFTWARE
0x981f9008 0x1b446008 \SystemRoot\System32\Config\DEFAULT
0x98209518 0x1f3a7518 \SystemRoot\System32\Config\SAM
0x982d55c0 0x1e17c5c0 \REGISTRY\MACHINE\BCD00000000
0xa620b008 0x1997b008 \??\C:\Users\IEUser\AppData\Local\Microsoft\Windows\UsrClass.dat
0xa65269c8 0x17ad09c8 \??\C:\Users\sshd_server\ntuser.dat
0xa6539260 0x0f690260 \??\C:\Users\sshd_server\AppData\Local\Microsoft\Windows\UsrClass.dat
0xa7d2a9c8 0x01c469c8 \??\C:\System Volume Information\Syscache.hve

┌──(icdfa㉿icdfa)-[~/CASE3]
└─$
```

UNDERSTANDING THE SUSPECT'S PC

SCREENSHOT 7 (EVIDENCE 7)



```
┌──(icdfa㉿icdfa)-[~/CASE3]
└─$ vol.py -f memdumpWin7.mem --profile=Win7SP1x86_23418 hivelist
Volatility Foundation Volatility Framework 2.6.1
Virtual    Physical   Name
---------- ---------- ----
0x87c10370 0x27fd5370 [no name]
0x87c1c008 0x27fa3008 \REGISTRY\MACHINE\SYSTEM
0x87c459c8 0x27d8e9c8 \REGISTRY\MACHINE\HARDWARE
0x889c0430 0x1285a430 \??\C:\Windows\ServiceProfiles\NetworkService\NTUSER.DAT
0x8b46d008 0x16d8c008 \??\C:\Windows\ServiceProfiles\LocalService\NTUSER.DAT
0x8b5d39b0 0x120049b0 \??\C:\Users\IEUser\ntuser.dat
0x97104008 0x1c45f008 \SystemRoot\System32\Config\SECURITY
0x9711d9c8 0x1e6a09c8 \SystemRoot\System32\Config\SOFTWARE
0x981f9008 0x1b446008 \SystemRoot\System32\Config\DEFAULT
0x98209518 0x1f3a7518 \SystemRoot\System32\Config\SAM
0x982d55c0 0x1e17c5c0 \REGISTRY\MACHINE\BCD00000000
0xa620b008 0x1997b008 \??\C:\Users\IEUser\AppData\Local\Microsoft\Windows\UsrClass.dat
0xa65269c8 0x17ad09c8 \??\C:\Users\sshd_server\ntuser.dat
0xa6539260 0x0f690260 \??\C:\Users\sshd_server\AppData\Local\Microsoft\Windows\UsrClass.dat
0xa7d2a9c8 0x01c469c8 \??\C:\System Volume Information\Syscache.hve

┌──(icdfa㉿icdfa)-[~/CASE3]
└─$
```

SCREENSHOT 8 (EVIDENCE 8)



```
┌──(icdfa㉿icdfa)-[~/CASE3]
└─$ vol.py -f memdumpWin7.mem --profile=Win7SP1x86_23418 printkey -o 0x87c459c8 -K  "DESCRIPTION\System\CentralProcessor\0"
Volatility Foundation Volatility Framework 2.6.1
Legend: (S) = Stable   (V) = Volatile

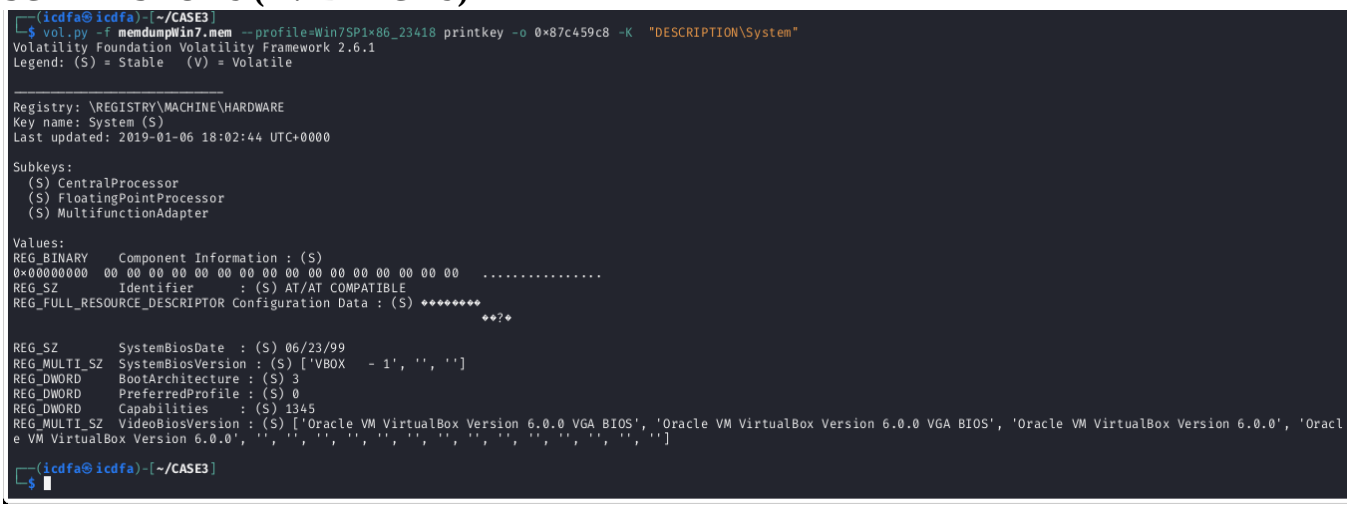
----------------------------
Registry: \REGISTRY\MACHINE\HARDWARE
Key name: 0 (S)
Last updated: 2019-01-06 18:02:40 UTC+0000

Subkeys:

Values:
REG_BINARY    Component Information : (S)
0x00000000  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00   ................
REG_SZ        Identifier      : (S) x86 Family 6 Model 158 Stepping 9
REG_FULL_RESOURCE_DESCRIPTOR Configuration Data : (S) ????????
REG_SZ        ProcessorNameString : (S) Intel(R) Core(TM) i7-7700 CPU @ 3.60GHz
REG_SZ        VendorIdentifier : (S) GenuineIntel
REG_DWORD     FeatureSet      : (S) 2697805823
REG_DWORD     ~MHz            : (S) 3600
REG_DWORD     Update Status   : (S) 2
REG_BINARY    Update Signature : (S)
0x00000000  00 00 00 00 00 00 00 00                         ........
REG_BINARY    Previous Update Signature : (S)
0x00000000  00 00 00 00 00 00 00 00                         ........
REG_DWORD     Platform ID     : (S) 2

┌──(icdfa㉿icdfa)-[~/CASE3]
└─$
```

SCREENSHOT 9 (EVIDENCE 9)



```
┌──(icdfa㉿icdfa)-[~/CASE3]
└─$ vol.py -f memdumpWin7.mem --profile=Win7SP1x86_23418 printkey -o 0x87c459c8 -K  "DESCRIPTION\System"
Volatility Foundation Volatility Framework 2.6.1
Legend: (S) = Stable   (V) = Volatile

----------------------------
Registry: \REGISTRY\MACHINE\HARDWARE
Key name: System (S)
Last updated: 2019-01-06 18:02:44 UTC+0000

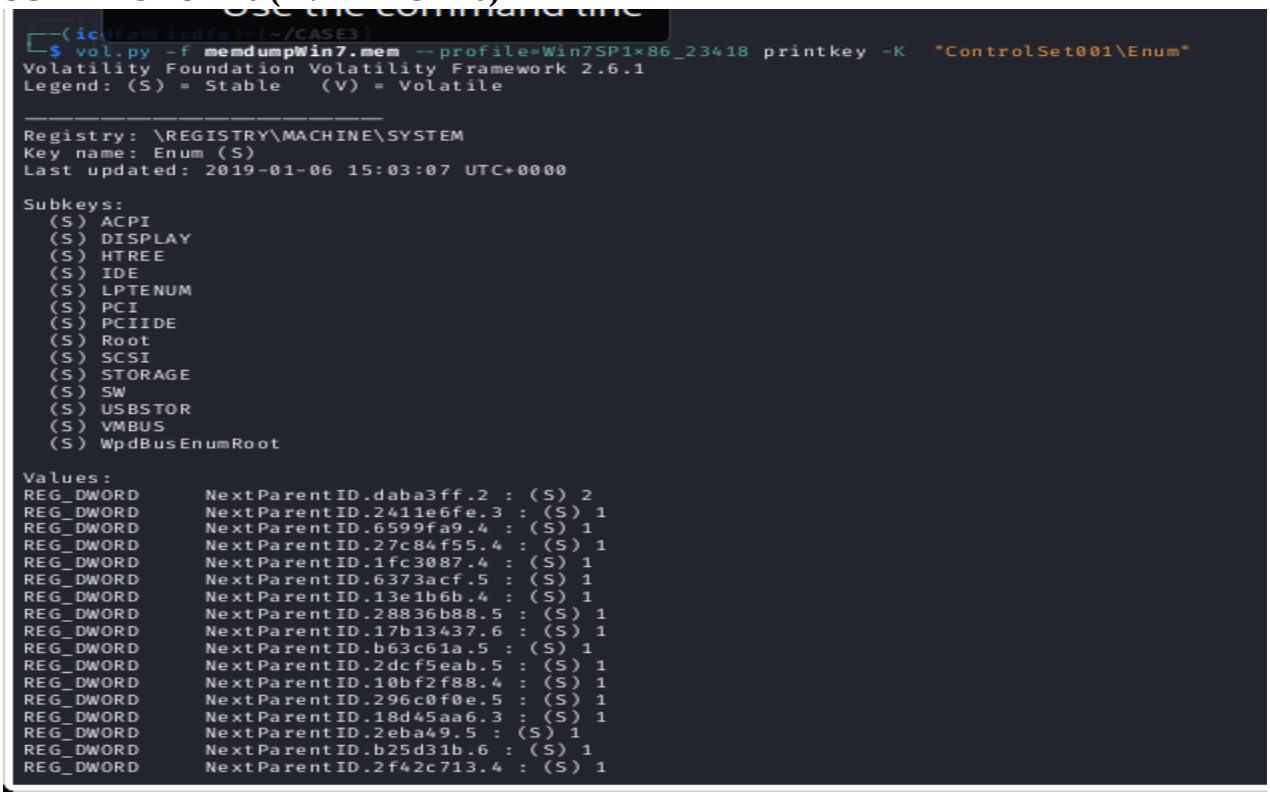
Subkeys:
  (S) CentralProcessor
  (S) FloatingPointProcessor
  (S) MultifunctionAdapter

Values:
REG_BINARY    Component Information : (S)
0x00000000  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00   ................
REG_SZ        Identifier      : (S) AT/AT COMPATIBLE
REG_FULL_RESOURCE_DESCRIPTOR Configuration Data : (S) ♦♦♦♦♦♦♦♦
                                                      ♦♦?♦

REG_SZ        SystemBiosDate  : (S) 06/23/99
REG_MULTI_SZ  SystemBiosVersion : (S) ['VBOX   - 1', '', '']
REG_DWORD     BootArchitecture : (S) 3
REG_DWORD     PreferredProfile : (S) 0
REG_DWORD     Capabilities    : (S) 1345
REG_MULTI_SZ  VideoBiosVersion : (S) ['Oracle VM VirtualBox Version 6.0.0 VGA BIOS', 'Oracle VM VirtualBox Version 6.0.0 VGA BIOS', 'Oracle VM VirtualBox Version 6.0.0', 'Oracl
e VM VirtualBox Version 6.0.0', '', '', '', '', '', '', '', '', '', '', '', '', '', '', '', '', '']

┌──(icdfa㉿icdfa)-[~/CASE3]
└─$
```

SCREENSHOT 10 (EVIDENCE 10)



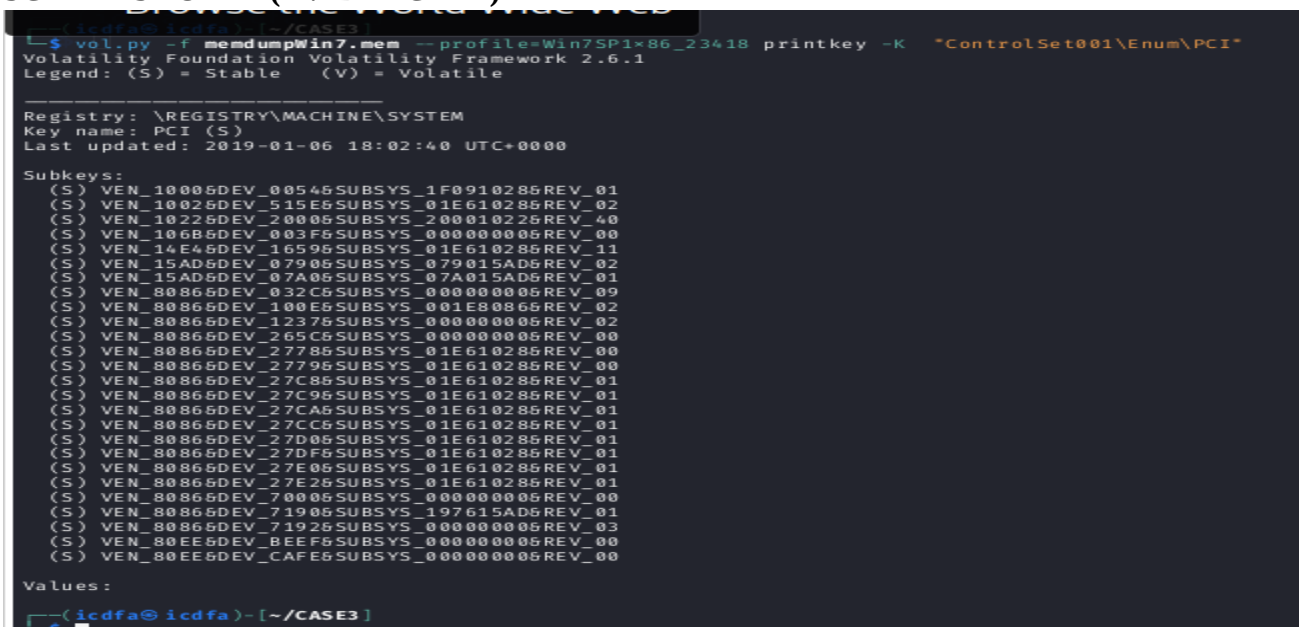
```
┌──(icdfa㉿icdfa)-[~/CASE3]
└─$ vol.py -f memdumpWin7.mem --profile=Win7SP1x86_23418 printkey -K  "ControlSet001\Enum"
Volatility Foundation Volatility Framework 2.6.1
Legend: (S) = Stable   (V) = Volatile

----------------------------
Registry: \REGISTRY\MACHINE\SYSTEM
Key name: Enum (S)
Last updated: 2019-01-06 15:03:07 UTC+0000

Subkeys:
  (S) ACPI
  (S) DISPLAY
  (S) HTREE
  (S) IDE
  (S) LPTENUM
  (S) PCI
  (S) PCIIDE
  (S) Root
  (S) SCSI
  (S) STORAGE
  (S) SW
  (S) USBSTOR
  (S) VMBUS
  (S) WpdBusEnumRoot

Values:
REG_DWORD     NextParentID.daba3ff.2 : (S) 2
REG_DWORD     NextParentID.2411e6fe.3 : (S) 1
REG_DWORD     NextParentID.6599fa9.4 : (S) 1
REG_DWORD     NextParentID.27c84f55.4 : (S) 1
REG_DWORD     NextParentID.1fc3087.4 : (S) 1
REG_DWORD     NextParentID.6373acf.5 : (S) 1
REG_DWORD     NextParentID.13e1b6b.4 : (S) 1
REG_DWORD     NextParentID.28836b88.5 : (S) 1
REG_DWORD     NextParentID.17b13437.6 : (S) 1
REG_DWORD     NextParentID.b63c61a.5 : (S) 1
REG_DWORD     NextParentID.2dcf5eab.5 : (S) 1
REG_DWORD     NextParentID.10bf2f88.4 : (S) 1
REG_DWORD     NextParentID.296c0f0e.5 : (S) 1
REG_DWORD     NextParentID.18d45aa6.3 : (S) 1
REG_DWORD     NextParentID.2eba49.5 : (S) 1
REG_DWORD     NextParentID.b25d31b.6 : (S) 1
REG_DWORD     NextParentID.2f42c713.4 : (S) 1
```

SCREENSHOT 11 (EVIDENCE 11)



```
└─$ vol.py -f memdumpWin7.mem --profile=Win7SP1x86_23418 printkey -K "ControlSet001\Enum\PCI"
Volatility Foundation Volatility Framework 2.6.1
Legend: (S) = Stable   (V) = Volatile

----------------------------
Registry: \REGISTRY\MACHINE\SYSTEM
Key name: PCI (S)
Last updated: 2019-01-06 18:02:40 UTC+0000

Subkeys:
  (S) VEN_1000&DEV_0054&SUBSYS_1F091028&REV_01
  (S) VEN_1002&DEV_515E&SUBSYS_01E61028&REV_02
  (S) VEN_1022&DEV_2000&SUBSYS_20001022&REV_40
  (S) VEN_106B&DEV_003F&SUBSYS_00000000&REV_00
  (S) VEN_14E4&DEV_1659&SUBSYS_01E61028&REV_11
  (S) VEN_15AD&DEV_0790&SUBSYS_079015AD&REV_02
  (S) VEN_15AD&DEV_07A0&SUBSYS_07A015AD&REV_01
  (S) VEN_8086&DEV_032C&SUBSYS_00000000&REV_09
  (S) VEN_8086&DEV_100E&SUBSYS_001E8086&REV_02
  (S) VEN_8086&DEV_1237&SUBSYS_00000000&REV_02
  (S) VEN_8086&DEV_265C&SUBSYS_00000000&REV_00
  (S) VEN_8086&DEV_2778&SUBSYS_01E61028&REV_00
  (S) VEN_8086&DEV_2779&SUBSYS_01E61028&REV_00
  (S) VEN_8086&DEV_27C8&SUBSYS_01E61028&REV_01
  (S) VEN_8086&DEV_27C9&SUBSYS_01E61028&REV_01
  (S) VEN_8086&DEV_27CA&SUBSYS_01E61028&REV_01
  (S) VEN_8086&DEV_27CC&SUBSYS_01E61028&REV_01
  (S) VEN_8086&DEV_27D0&SUBSYS_01E61028&REV_01
  (S) VEN_8086&DEV_27DF&SUBSYS_01E61028&REV_01
  (S) VEN_8086&DEV_27E0&SUBSYS_01E61028&REV_01
  (S) VEN_8086&DEV_27E2&SUBSYS_01E61028&REV_01
  (S) VEN_8086&DEV_7000&SUBSYS_00000000&REV_00
  (S) VEN_8086&DEV_7190&SUBSYS_197615AD&REV_01
  (S) VEN_8086&DEV_7192&SUBSYS_00000000&REV_03
  (S) VEN_80EE&DEV_BEEF&SUBSYS_00000000&REV_00
  (S) VEN_80EE&DEV_CAFE&SUBSYS_00000000&REV_00

Values:

┌──(icdfa㉿icdfa)-[~/CASE3]
```

SCREENSHOT 12 (EVIDENCE 12)



```
┌──(icdfa㉿icdfa)-[~/CASE3]
└─$ vol.py -f memdumpWin7.mem --profile=Win7SP1x86_23418 printkey -K "ControlSet001\Control\ComputerName"

Volatility Foundation Volatility Framework 2.6.1
Legend: (S) = Stable   (V) = Volatile

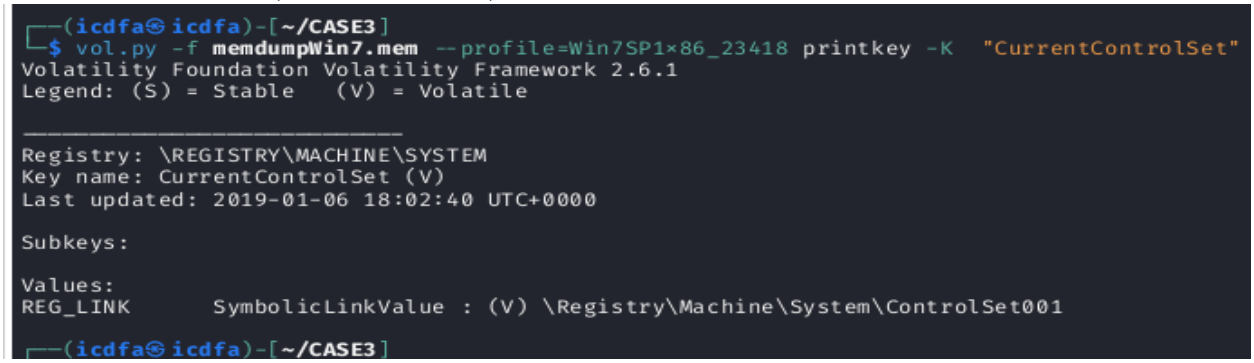
-------------------------
Registry: \REGISTRY\MACHINE\SYSTEM
Key name: ComputerName (S)
Last updated: 2019-01-06 18:02:44 UTC+0000

Subkeys:
  (S) ComputerName
  (V) ActiveComputerName

Values:

┌──(icdfa㉿icdfa)-[~/CASE3]
└─$
```

SCREENSHOT 13 (EVIDENCE 13)



```
┌──(icdfa㉿icdfa)-[~/CASE3]
└─$ vol.py -f memdumpWin7.mem --profile=Win7SP1x86_23418 printkey -K "CurrentControlSet"
Volatility Foundation Volatility Framework 2.6.1
Legend: (S) = Stable   (V) = Volatile

---------------------------
Registry: \REGISTRY\MACHINE\SYSTEM
Key name: CurrentControlSet (V)
Last updated: 2019-01-06 18:02:40 UTC+0000

Subkeys:

Values:
REG_LINK      SymbolicLinkValue : (V) \Registry\Machine\System\ControlSet001

┌──(icdfa㉿icdfa)-[~/CASE3]
```

NETWORK FORENSICS

SCREENSHOT 14 (EVIDENCE 14)

```
┌──(kali㉿kali)-[~/volatility3]
└─$ python3 vol.py -f memdumpWin7.mem windows.netscan | grep TCPv4
0×21478bb0 100.0TCPv4    0.0.0.0 49155   0.0.0.0 0       LISTENING       432     services.exe    N/A
0×269ff708      TCPv4    0.0.0.0 49156   0.0.0.0 0       LISTENING       440     lsass.exe       N/A
0×3e043b10      TCPv4    192.168.56.8    49177   192.168.56.5    80      CLOSED  -       -       -
0×3e446008      TCPv4    0.0.0.0 49154   0.0.0.0 0       LISTENING       880     svchost.exe     N/A
0×3e446d98      TCPv4    0.0.0.0 49154   0.0.0.0 0       LISTENING       880     svchost.exe     N/A
0×3e46e0c0      TCPv4    192.168.56.8    139     0.0.0.0 0       LISTENING       4       System  N/A
0×3e671988      TCPv4    0.0.0.0 445     0.0.0.0 0       LISTENING       4       System  N/A
0×3e72e530      TCPv4    0.0.0.0 135     0.0.0.0 0       LISTENING       672     svchost.exe     N/A
0×3e72ee48      TCPv4    0.0.0.0 135     0.0.0.0 0       LISTENING       672     svchost.exe     N/A
0×3e736388      TCPv4    0.0.0.0 49152   0.0.0.0 0       LISTENING       340     wininit.exe     N/A
0×3e736d98      TCPv4    0.0.0.0 49152   0.0.0.0 0       LISTENING       340     wininit.exe     N/A
0×3e79d528      TCPv4    0.0.0.0 49153   0.0.0.0 0       LISTENING       724     svchost.exe     N/A
0×3e7a1618      TCPv4    0.0.0.0 49153   0.0.0.0 0       LISTENING       724     svchost.exe     N/A
0×3e8f0620      TCPv4    0.0.0.0 49156   0.0.0.0 0       LISTENING       440     lsass.exe       N/A
0×3e9b3e50      TCPv4    0.0.0.0 49155   0.0.0.0 0       LISTENING       432     services.exe    N/A
0×3ee1aa88      TCPv4    0.0.0.0 22      0.0.0.0 0       LISTENING       2016    sshd.exe        N/A
0×3ee1bc18      TCPv4    0.0.0.0 22      0.0.0.0 0       LISTENING       2016    sshd.exe        N/A
0×3fcba008      TCPv4    192.168.56.8    49178   192.168.56.5    80      CLOSED  -       -       -

┌──(kali㉿kali)-[~/volatility3]
```

SCREENSHOT 15 (EVIDENCE 15)

```
┌──(kali㉿kali)-[~/volatility3]
└─$ python3 vol.py -f memdumpWin7.mem windows.pslist
Volatility 3 Framework 2.28.3
Progress:  100.00               PDB scanning finished
PID     PPID    ImageFileName   Offset(V)       Threads Handles SessionId       Wow64   CreateTime      ExitTime        File output

4       0       System  0×8413a908      73      518     N/A     False   2019-01-06 18:02:44.000000 UTC  N/A     Disabled
236     4       smss.exe        0×851ec488      2       29      N/A     False   2019-01-06 18:02:44.000000 UTC  N/A     Disabled
300     292     csrss.exe       0×852b11e0      8       434     0       False   2019-01-06 18:02:44.000000 UTC  N/A     Disabled
340     292     wininit.exe     0×841c32b8      3       75      0       False   2019-01-06 18:02:44.000000 UTC  N/A     Disabled
348     328     csrss.exe       0×852b21d0      9       316     1       False   2019-01-06 18:02:44.000000 UTC  N/A     Disabled
388     328     winlogon.exe    0×858a4d28      3       112     1       False   2019-01-06 18:02:45.000000 UTC  N/A     Disabled
```

INVESTIGATING THE SUSPECT COMMAND HISTORY

SCREENSHOT 16 (EVIDENCE 16)

```
3784    432     svchost.exe     0×858be710      14      370     0       False   2019-01-06 15:04:51.000000 UTC  N/A     Disabled
3996    2524    cmd.exe 0×8431fb08      1       22      1       False   2019-01-06 15:06:11.000000 UTC  N/A     Disabled
3988    348     conhost.exe     0×842ddd28      2       52      1       False   2019-01-06 15:06:11.000000 UTC  N/A     Disabled
976     2388    chrome.exe      0×842eb6b0      14      198     1       False   2019-01-06 15:06:23.000000 UTC  N/A     Disabled
2596    2524    FTK Imager.exe  0×84312030      14      358     1       False   2019-01-06 15:07:15.000000 UTC  N/A     Disabled
2228    552     dllhost.exe     0×850d5a38      6       1702389038      1       False   2019-01-06 15:09:07.000000 UTC  N/A     Disabled

┌──(kali㉿kali)-[~/volatility3]
```

INVESTIGATING THE SUSPECT'S USB

SCREENSHOT 18 (EVIDENCE 18)



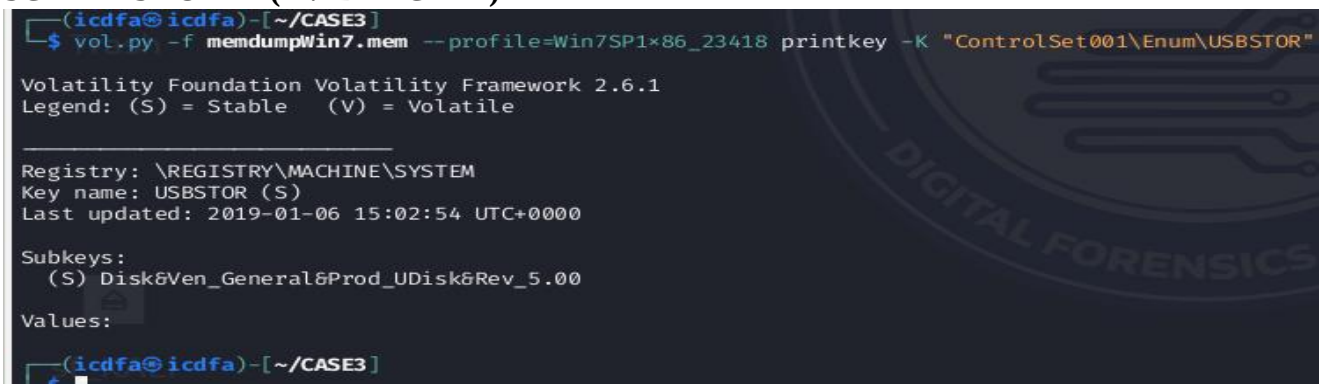
SCREENSHOT 19 (EVIDENCE 19)



SCREENSHOT 20 (EVIDENCE 20)



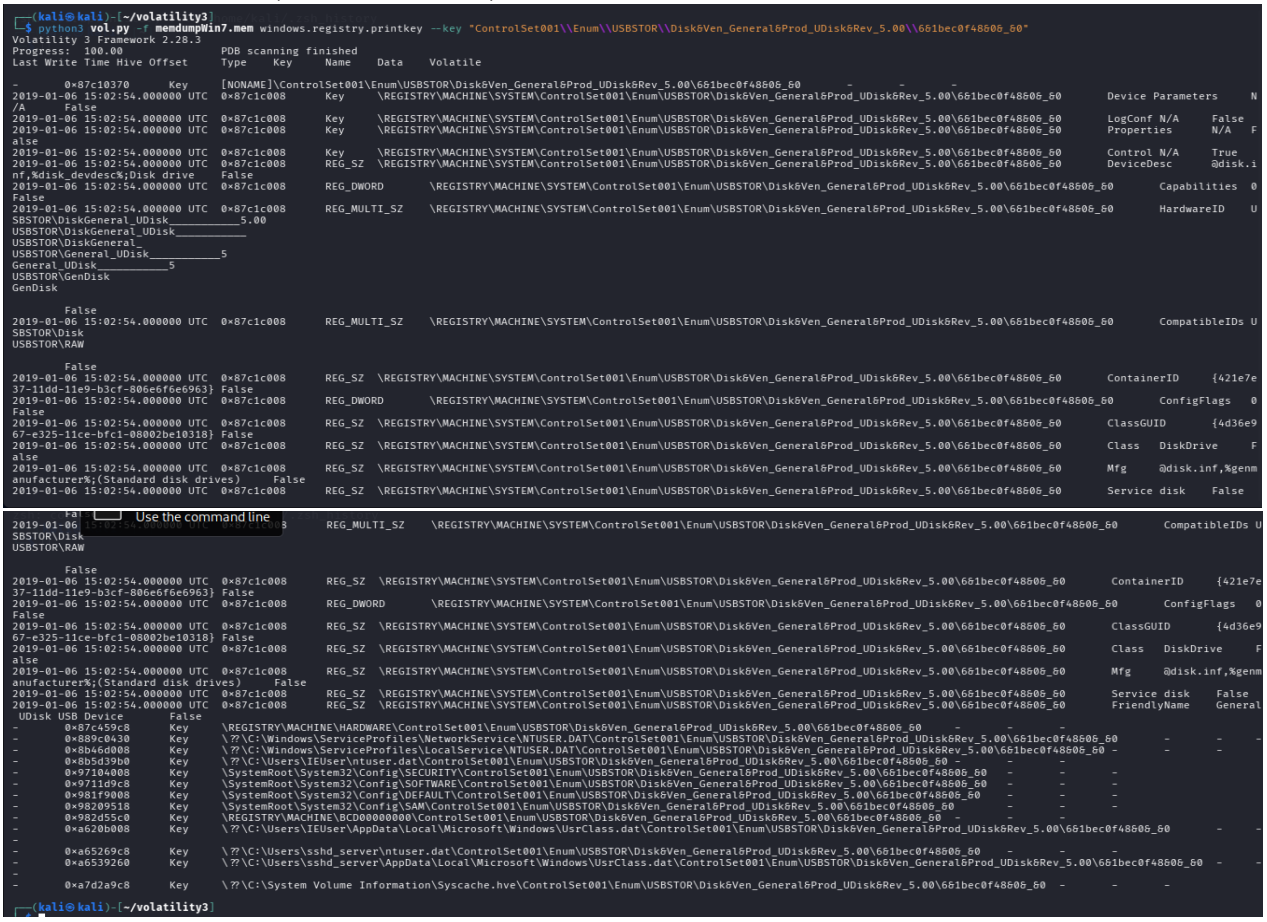
SCREENSHOT 21 (EVIDENCE 21)



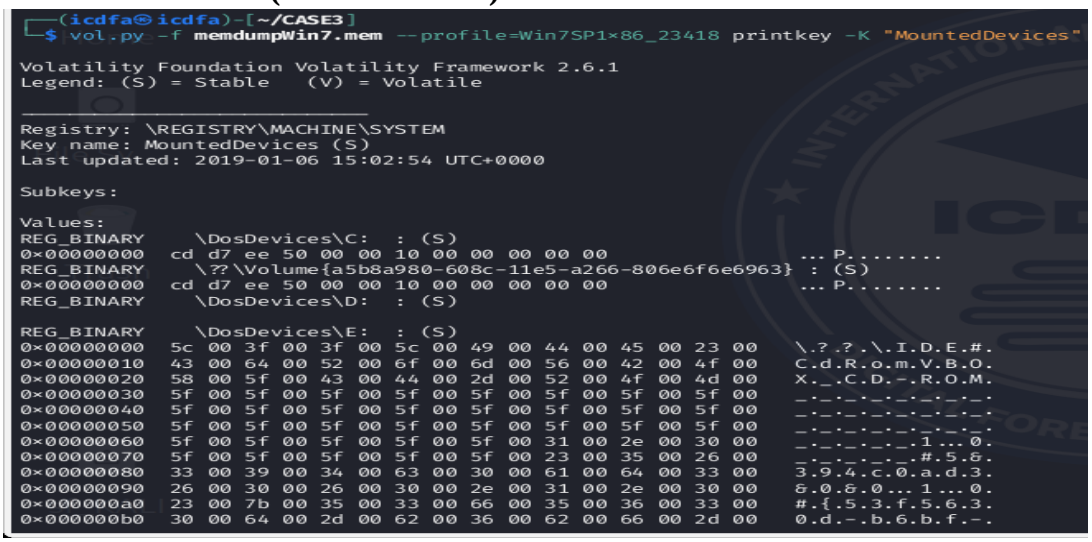
SCREENSHOT 22 (EVIDENCE 22)



SCREENSHOT 23 (EVIDENCE 23)



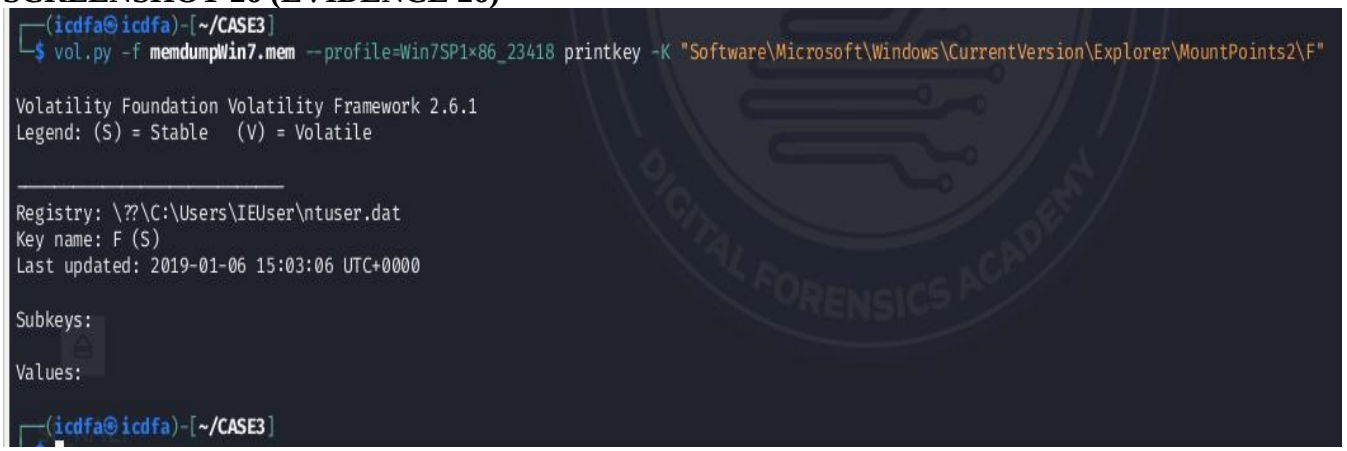
SCREENSHOT 24 (EVIDENCE 24)



SCREENSHOT 25 (EVIDENCE 25)

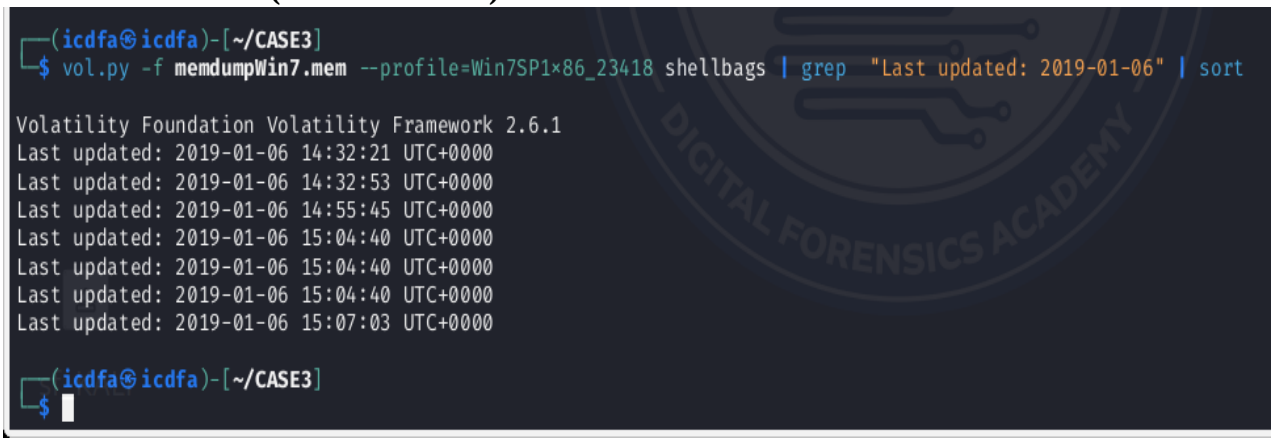


SCREENSHOT 26 (EVIDENCE 26)

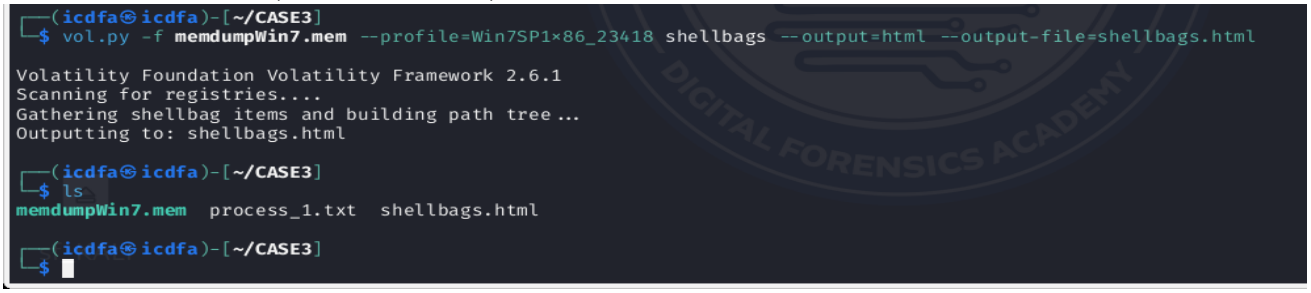


INVESTIGATING THE SUSPECT FILE EXPLORER HISTORY

SCREENSHOT 28 (EVIDENCE 28)



SCREENSHOT 29 (EVIDENCE 29)



Analysis and Discussion

The below are my finding based on the various aspect of the memory forensic analysis

1) Understanding the Suspect and Accounts**:-** while analysis to understand the suspect and his account I was able to find out that the suspect user name is **IEUser** whch is also the default logon user.

2) Understanding the Suspect's PC:- in understanding the suspect's PC, the suspect uses a PC that has the following properties (Intel(R) core(TM) i7-700, CPU@3.60GHZ) the suspect computer name/device name is IE8WIN7 and also some list of devices that connect to the suspect device include ACPI,DISPLAY.

3) Network Forensics:- during my analysis in the network aspect of the memory file two ip addresses one local ip address(192.168.56.5) and a foreign ip address (192.168.56.8) were visible each using a different port number for communication.

4) Investigating the Suspect's USB:-while investigating the suspect USB, many drive interface ID were visible but the suspect USB which has an interface ID(53f56307-b6bf-11d0-94f2-00a0c91efb8b) was identified from the from the list of different drive interface ID. The suspect USB had a instance ID of (6&1bec0f48&0&_&0) which belong to a class of (DiskDrive) and a classGUID of(4d36e967-e325-11ce-bfc1-08002be10318) and this same USB was last plugged into the system on 06-01-2019 @ 15:03:06

5) Investigating the File Explorer History:- in regarding the file explorer history, the download folder with the path(C/users/IEUser/Downloads) was updated on the same date(2019-01-06) but at different time interval which means either the suspect has downloaded a file into the download folder or has delete a file from the download folder.

Limitation

The limitation/challenges I faced during the analyses is that not all I executed produce an output and I have tried several alternative such as installing volatility but still I couldn't get any result which is the result why I couldn't get all the evidence/screenshot completed.

Conclusion

In conclusion this lab has provide me with me the basic knowledge in understanding how to use the volatility framework to analyse a memory image.

Reference

➢ICDFA course Material (SBT-DF204(computer forensic case studies) by COMMANDANT AMINU IDRIS